

Math 125: Quiz 1

Name: _____

NetID: _____

Instructions. Show all work. Answers without supporting work may receive little or no credit.

1. Determine whether the following limit exists. If it exists, compute it.

[5 points]

$$\lim_{x \rightarrow 0} \frac{\sqrt{9+x} - 3}{x}.$$

2. Use the Squeeze Theorem to compute the following limit from the right. Clearly state the inequalities you use.

[5 points]

$$\lim_{x \rightarrow 0^+} x^2 \sin\left(\frac{1}{x}\right).$$